

PBPK-Guided Generative Drug Design: From Static Molecular Properties to Time-Resolved In Vivo Optimization

Project concept for a Marie Skłodowska-Curie Global Postdoctoral Fellowship

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Background and rationale

Generative AI is increasingly used to design novel drug candidates, optimizing molecules against predefined potency, physicochemical and safety properties. Current workflows, however, rely on static molecular descriptors and computational proxies that do not directly capture in vivo behavior -- absorption, distribution, metabolism, and the resulting concentration-time profiles. Physiologically based pharmacokinetic (PBPK) modeling offers a mechanistic framework to close this gap, but is currently used almost exclusively as a downstream evaluation step, after candidates have already been generated.

The central hypothesis of this project is that mechanistic pharmacokinetic information can instead be incorporated directly into generative molecular optimization, enabling the design of compounds that are not only potent and chemically feasible, but also predicted to achieve desirable in-vivo exposure and safety profiles.

Preliminary work at ICSI Berkeley

During my current DAAD-funded postdoctoral stay at ICSI, I tested whether conventional evaluation criteria for generative drug-design models adequately capture pharmacokinetic and safety properties. Using a PBPK simulation layer, I evaluated candidates from three independent generative pipelines, all already pre-filtered to a favorable static ADME/toxicity profile. The results point to systematic gaps: a meaningful share of these pre-filtered candidates turn out to have no pharmacokinetically achievable dose at all, roughly 70% cannot be confidently classified as safe once prediction and simulation uncertainty is propagated, and about 85% are each simultaneously outperformed on potency and safety by an alternative that conventional scoring did not select.

This preliminary audit indicates that conventional evaluation criteria do not reliably identify pharmacokinetically favorable candidates. The proposed project builds directly on this finding, as detailed for each phase below.

Research objectives

The proposed MSCA project will develop a PBPK-guided generative drug-design framework in which mechanistic pharmacokinetic predictions become an explicit component of candidate generation and optimization, pursued through three objectives:

Objective 1 - A PBPK-aware molecular optimization framework

Connect generative molecular models to mechanistic PBPK simulation via a trainable surrogate, so that predicted pharmacokinetic exposure directly guides molecule generation and optimization -- rather than being applied as a post hoc filter on already-generated candidates, as in the preliminary DAAD-funded audit.

Objective 2 - Time-resolved pharmacokinetic optimization

Rather than optimizing a single predicted PK parameter, use concentration-time profiles and

exposure-related characteristics as optimization objectives, to identify molecules with desirable in-vivo exposure while maintaining potency and safety. This will be developed jointly with Pavia's population PK/PD expertise to incorporate inter-individual variability into the exposure objective -- a limitation of the preliminary DAAD work, which relied on a single virtual individual.

Objective 3 - Generalizability and biological relevance

Evaluate the resulting framework across molecular generation strategies and drug-design tasks, with emphasis on uncertainty, potency/safety/PK trade-offs, and how PBPK-guided optimization changes the composition and predicted quality of generated candidate sets.

Research environment and knowledge transfer

ICSI Berkeley - 18 months

The DAAD-funded audit evaluated candidates from fixed, already-existing generative pipelines after the fact. The MSCA outgoing phase goes further: predicted pharmacokinetic exposure becomes an active part of the generative process itself, via a trainable surrogate that closes the loop between molecular design and PBPK simulation, rather than scoring already-generated output afterward.

University of Pavia - 12 months

The return phase adds a dimension the ICSI-based preliminary work explicitly lacked: population-level pharmacokinetic variability. The DAAD-funded audit relied on a single virtual individual and did not model inter-individual variability; Professor Paolo Magni's expertise in population PK/PD and nonlinear mixed-effects modeling will be used to incorporate this variability directly into the time-resolved optimization objective, rather than as a general pharmacometrics consultation. This establishes a bidirectional knowledge transfer in which Pavia contributes a specific methodological capability the outgoing phase alone cannot provide.

Expected outcomes and impact

The project aims to move generative drug design from static property optimization toward in-vivo-aware molecular design. Expected outcomes include:

- a computational framework integrating generative AI with mechanistic PBPK modeling, using time-resolved exposure information as an explicit optimization signal;
- a systematic assessment of where conventional potency/safety objectives fail to predict PBPK-confirmed outcomes; and
- publications and open, reproducible workflows at the intersection of generative AI, computational drug discovery and quantitative pharmacology.

In the longer term, the project could help establish AI-driven drug design in which pharmacokinetic behavior is considered alongside potency and safety from the start, reducing the number of computationally promising but pharmacokinetically unsuitable candidates reaching later development stages.